VAGRANT HANDS-ON

Commands

- Vagrant add
- vagrant up
- vagrant status
- vagrant suspend
- vagrant resume
- vagrant halt
- vagrant destroy
- vagrant snapshot
- vagrant box list
- vagrant box add
- vagrant box outdated
- vagrant box remove
- vagrant package

Before running these commands, ensure:

You have Vagrant installed

You have VirtualBox installed (or another supported provider)

You have an initialized Vagrantfile:

```
vagrant init hashicorp/bionic64
```

The name of the Vagrant box (Ubuntu 18.04 LTS, 64-bit) from Vagrant Cloud

vagrant up — Bring a box online

Explanation:

- Downloads the box if not already installed
- Starts the VM using the provider (e.g., VirtualBox)
- Runs any provisioning script

```
vagrant init ubuntu/bionic64
vagrant up
```

vagrant status — Show current box status

Example output:

Current machine states:
default running (virtualbox)

vagrant suspend — Pause the current box

Saves the current state to memory and suspends the VM (like sleep mode).

Use case: Save state quickly without shutting down.

vagrant resume — Resume the current box

Resumes a suspended VM.

vagrant halt — Shutdown the current box

Gracefully shuts down the VM.

vagrant halt

Use case: To save resources without destroying your environment.

vagrant halt and vagrant suspend stop a running virtual machine, but they do it differently — ***with different effects on performance, saved state, and resume time.***

vagrant halt — Graceful Shutdown

- **What it does:** Performs a clean shutdown (like powering off a physical machine).
- **How it works:** Uses the operating system's shutdown process.
- **Resume time:** Starts from scratch (boot time).
- **RAM/CPU freed:** Yes (completely).
- **Data loss:** No (unless unsaved changes were in memory).

When to Use:

- When you're done for the day.
- When you want to conserve all system resources.
- Before packaging the machine with vagrant package.

vagrant suspend — Save to Memory (Sleep Mode)

- **What it does:** Pauses the VM and saves its entire memory and CPU state to disk.
- **How it works:** Like putting a physical machine to sleep.
- **Resume time:** Very fast (picks up exactly where you left off).
- **RAM/CPU freed:** Yes (but disk usage increases).
- **Data loss:** No — everything in memory is restored.

When to Use:

- Short breaks.
- You want to resume work instantly without rebooting the OS inside VM.
- Testing or debugging environments.

vagrant destroy — Destroy the current box

- This will **remove the VM** and any data not saved externally.

vagrant snapshot — Save/restore state of VM

Take a snapshot:

vagrant snapshot save my_snapshot

vagrant snapshot list:

vagrant snapshot list

Restore snapshot:

vagrant snapshot restore my_snapshot

Delete snapshot:

vagrant snapshot delete my_snapshot

vagrant box list — List installed boxes

vagrant box list

Example output:

ubuntu/bionic64 (virtualbox, 20201211.0.0)

vagrant box add — Download a box

Adds a new box manually from Vagrant Cloud or a file.
CLOUD: If you're downloading a box from Vagrant Cloud, HashiCorp's public repository of Vagrant boxes
From cloud:
vagrant box add ubuntu/focal64

LOCAL: If you're adding a box that is already saved as a .box file on your system (e.g., custom, exported, or offline box).
From local:
vagrant box add my_custom_box file://path/to/mybox.box

vagrant box outdated — Check if box has updates

Compares installed box version with the cloud version.

(i.e., if you're using a cloud-based box like ubuntu/focal64).

vagrant box update — Update the box

Downloads the latest version of the box.

RECALL

vagrant destroy — Destroys the VM (instance)

- **What it does:** Deletes the **virtual machine instance** that was created with vagrant up.
- **Effect on disk:** Deletes the actual running VM (removes it from VirtualBox or other provider).
- **Box file? ✅ Keeps** the base box you originally downloaded

vagrant box remove — Deletes the base box image

- **What it does:** Removes the **downloaded base box image** stored locally on your computer.
- **Effect on disk:** Frees up disk space by deleting the .box file stored in ~/.vagrant.d/boxes/.
- **VMs using it?** ❌ Will not affect any running machines (they may break if you try to vagrant up again).
- **Command:**

```
vagrant box remove ubuntu/focal64
```

Action	vagrant destroy	vagrant box remove
Affects running VM?	Yes — deletes it	No — existing VMs keep working
Affects base image?	No — base .box still available	Yes — deletes the .box file
Frees up storage?	Some (VM space)	Yes (removes base image)
Reversible?	Can recreate with vagrant up	You must re-download the box again

vagrant package — Create a custom box

Packages a running VM into a .box file.

```
vagrant package --output my_custom.box
```

This command exports your current VM into a portable .box file so that you (or others) can later use it with vagrant box add.

EXAMPLE

```
# Vagrantfile
Vagrant.configure("2") do |config|
  config.vm.box = "ubuntu/bionic64"

  config.vm.define "web_server" do |web_server|
    web_server.vm.hostname = "web-server"
    web_server.vm.network "private_network", type: "dhcp", ip: "192.168.33.10"

    web_server.vm.provider "virtualbox" do |vb|
      vb.memory = "512"
      vb.cpus = 1
    end
  end
end
```

```
Vagrant.configure("2") do |config|
```

- Starts the Vagrant configuration block.
- "2" refers to Vagrant configuration version 2 (used for most modern setups).
- config is the main configuration object.

config.vm.box = "ubuntu/bionic64"

- Specifies the base box to use: in this case, **Ubuntu 18.04 LTS**.
- It will be downloaded from Vagrant Cloud if not already on your system.

config.vm.define "web_server" do |web_server|

- Creates a **named VM** called "web_server".
- Allows you to define multiple VMs in the same Vagrantfile if needed.
- web_server is a local variable for configuring this specific VM.

web_server.vm.hostname = "web-server"

•Sets the internal hostname of the VM to web-server.

•This is the name the machine will report inside the guest OS.

web_server.vm.network "private_network", type: "dhcp", ip: "192.168.33.10"

•Sets up private networking so the VM can communicate with the host.

•The VM gets an IP on a private network (here: 192.168.33.10), which is only accessible from your host.

web_server.vm.provider "virtualbox" do |vb|

•Configures provider-specific settings for VirtualBox.

•vb is a variable used to set properties related to the VirtualBox provider.